BSc in Computing / Artificial Intelligence

COMP2111 Database Design

Project: Group Work on DB Design (2026/2027)

Important notice:

- This project is for **a group of 4 or 5 students** to work together. Group size not following this requirement needs to obtain prior approval from module teacher.
- This project comprises 15% of the total mark of this module.
- **Submission of "Declaration on GenAI Usage" is required for each phase.**
- To deal with plagiarism, a zero mark will be given if you copy someone else's work or you let someone copy your work.
- **For each submission requirement, late submission will NOT be graded.** It's your responsibility to type my email address correctly, and submit the correct files that can be opened. No excuses on any of this for late submission.

Database Design Project

The objective of this group project is to apply the design and implementation concepts discussed in class throughout the semester. The project involves developing and documenting the conceptual, logical and physical database design, and implementing the design in Microsoft Access based on the **LibNet** case study (**read DB design project_Part2.pdf**). Table 1 shows the summary of this project by phases.

Table 1. Summary of database design project by phases

Phase	Description	Output	Due
Phase 0 **(5%)**	**Project Initiation** ○ Form project group	**Email** me the group list and cc copy to each member	Sept 21 (Monday) **I won't click on links to download**
	○ Plan your project	**Email** me a word file (cc copy to each member) –	Oct 5 (Monday) **I won't click on links to download**
Phase I: Ch. 16 **(30%)**	**Conceptual Database Design (follow details given)** ○ 3 data dictionaries ○ ER model of UML notation (can be drawn by hand)	Documentation to be **submitted in class**	Week of Oct 20 (**in-class**) (the first 20 minutes of official class time)
Phase II: Ch. 17 **(15%)**	**Logical Database Design (format MUST follow Ch. 17)** ○ Deriving Relations	Documentation to be **submitted in class**	Week of Nov 10 (**in-class**) (the first 20 minutes of official class time)
Phase III **(30%)**	**Physical Database Design / Implementation** ○ Create Tables ○ Input Data ○ Formulate the queries	**Email** me BOTH the Access database file & necessary documentation (cc copy to each member)	Nov 24 (Tuesday) **I won't click on links to download**
Peer Review **(20%)**	**Peer Review of Project** ○ Evaluate your team member	Peer Review Form	Nov 24 (Tuesday) **Via Canvas**

Stages of the Project

Phase 0 [5%]: Project Initiation

Step 1: Form Project Group

This is a group project. Students should form groups of 4 or 5. After you form the group, each group should email to **calanachan@mpu.edu.mo** (and c.c. to each group member) about members including name and student ID by **Sept 21 (Monday).** If I didn't receive any group member list email from you by the deadline, I will randomly form groups for you. **Subject of the Email: "COMP2111 DB Design – Project Group Members"**

Step 2: Formulate transaction requirements, considering EDI -- Oct 5 (Monday).

Study the user requirements of **LibNet** (**read DB design project_Part2.pdf)** carefully and then formulate **4 more cases** for each type of the transaction requirements, namely, data entry, date update / deletion, data queries. The queries must be realistic and you should only consider queries that need to retrieve data from **at least 2 different tables** to answer the request. Note, the queries that the database system is able to answer ultimately determine what information needs to be maintained in the database. In the last phase of the project, you need to write the SQL statements for these queries.

As you embark on this group project, consider it more than just a technical exercise—it's an opportunity to explore how database design can actively promote **equity, diversity, and inclusion (EDI).** By thinking critically about what data is collected, how it's structured, and whose identities and needs are represented, you'll begin to see how technology can either reinforce bias or challenge it. Use this project to reflect on the power of inclusive design: create systems that respect individual identities, accommodate diverse experiences, and support equitable access to resources. Your collaboration can spark meaningful conversations and inspire solutions that make digital spaces more just and representative for everyone.

[Submission guideline for Phase 0]

Complete the given "**DB design project_Phase0_template**" for submission, with a **SIGNED copy of "Declaration on GenAI Usage" – Email me the files as attachments, don't send me the links to download. Subject of the Email: "COMP2111 DB Design (Group No: xx) – Project Phase 0 submission"**

Phase I [30%]: Conceptual Database Design

Create a conceptual data model for **LibNet**. What you have submitted in Phase 0 only covers a subset of the data. You need to consider ALL the given data requirements for the 3 phases of Database Design.

Step 3: ER / EER modeling (Must be of UML notation)

Conceptual database design is an iterative process. Based on the notes of Ch. 16 on Conceptual Database Design, you have to show the step-by-step development (up to Step 1.6) of the conceptual data model supported by documentation, including **3** data dictionaries (ONE for entity – Step 1.1; ONE for relationship – Step 1.2; ONE for attributes – Step 1.3 to 1.5 combined together) and 1 ER diagram. **Include the attributes in your ER diagram**. It is NOT necessary to consider foreign keys at this stage, which will be handled in Phase 2.

Step 4: Specify ANY assumptions you have about the database in English, if necessary.
Do not make too many simplifying assumptions that are obvious.

[Submission guideline for Phase I]

A written report that includes student number and names of all members, with a **SIGNED copy of "Declaration on GenAI Usage"**. Phase I documentation should include sufficient detail to describe Steps 3 and 4. If you draw the ER model by hand, please take a copy of it as you need to refer to it in Phase 2.

Phase II [15%]: Logical Database Design

The objective of logical database design is to translate the conceptual data model into a logical data model by deriving a set of relations from the conceptual data model.

Step 5: Deriving Relations from conceptual data model

Convert the ER / EER diagram from Step 4 above into relations using the mapping steps discussed in Ch. 17. Make sure that the derived relations to be submitted are normalized to BCNF (**You don't have to submit normalization processes of this step**).

Make sure the derived relations to be submitted identifies **the primary key and foreign keys in all the relations, including the integrity constraints as well.**

[Submission guideline for Phase II]

A written report that includes student number and names of all members, with **a SIGNED copy of "Declaration on GenAI Usage"**. The body of Phase II documentation should include sufficient detail to describe the derived relational schema.

Phase III [30%]: Physical Database Design / Implementation

Use Microsoft Access to build the database tables from the normalized set of relations created in the previous phase, to meet **ALL** the given data requirements. Input some sample data for each table. Each table should contain at least **4** rows.

[Submission guideline for Phase III]

Create a compressed file (.zip or .rar) containing the followings and then send to my email address (calanachan@mpu.edu.mo). **Subject of the Email: "COMP2111 DB Design (Group No: xx) – Project Phase III submission".**

1. Microsoft Access database file (named using your group number) with the necessary data and your SQL queries (the 5 data queries submitted in phase 0);
2. A Microsoft Word file (named using your group number) **listing your SQL queries and the English description for your queries**, together with the screen capture of the result from running each of your queries.

You are responsible for the followings, or you will risk getting zero marks for this phase of the project:

- Typing my email address correctly when sending it to me (I would suggest you c.c. copy to yourself and each member to double check the correctness of my email address)
- You have attached the necessary files in the email.

Peer Evaluation [20%]: <u>Failing to submit peer evaluations will result in zero marks for your part on peer evaluation.</u>

The purpose of peer evaluation is to assess the contributions of each group member to complete the project. Rate each of the other group members in each category on a scale of 0 to 3. A rating of 3 represents that a member has contributed adequately and/or proportionately in that category. A rating of 0 is used to indicate that a group member did not contribute in any manner. If a group member receives a 0 from all of the other group members, that group member will not get any credit for the project. The peer evaluations impact 20% of entire project. The average value of your peer evaluation is used to calculate part of your total score.

Groups – fostering EDI

Group membership is the responsibility of the individual. To ensure fairness in grading, a peer evaluation component is required of everyone participating in a group for adjusting an individual's grade.

<u>Learning Objectives of the Group</u>

Group activities offer several invaluable professional development experiences. During this project, please be aware of the following cognitive activities you will be exposed to and develop yourself to become a better professional person:

- The *art* of listening (i.e., listening to what others have to say)
- Negotiation, persuasion, tolerance and humility (i.e., resolving differences and accepting the outcomes in a professional manner)
- Time management and coordination (i.e., working together through the assignment of tasks)
- Creative thought activity through parallel human processing (i.e., developing synergies through meaningful interactions among all group members)
- Leadership (i.e., taking responsibility for your actions and those of the group, promoting meaningful actions, setting an example for work actions and quality, bringing out the best among all group members)

As you work together on this project, remember that fostering equity, diversity, and inclusion begins with how you engage with one another. Every team member brings a unique perspective shaped by their background, experiences, and values—these differences are not just valid, they're vital. Respecting each other's ideas, listening actively, and creating space for all voices to be heard will enrich your collaboration and reflect the very principles of EDI you're aiming to promote. Inclusion starts in the conversation, so let curiosity, empathy, and openness guide your discussions.